

THE EVENT RULEBOOK (LEAVE ON SCREEN)

- **Soft Tire:** $T(x) = 78.00 + 0.15x^2$
- **Medium Tire:** $T(x) = 80.00 + 1.80x$
- **Hard Tire:** $T(x) = 82.00 + 0.60x$
- **Base Pit Stop Penalty:** 20.00s (Resets tire age x to 0)
- **The Traffic Jam Rule:** If 4 or more teams pit on the exact same lap, ALL pitting cars get a + 6.00s penalty (Total stop = 26.00s).

MOCK ROUND

- In website all teams log in and get familiar with the UI.

1. TEAM PAGE (Student Dashboard)

The Team Page needs two distinct UI states that toggle automatically based on the Admin's timer.

State A: Submission Phase (The 3-Minute Window)

- **Persistent Dashboard:**
 - Current total race time and overall leaderboard position (e.g., "P4").
 - Current Tire Compound (Soft, Medium, Hard).
 - Current Tire Age (x).
 - Reference card showing the exact degradation formulas and standard 20s pit penalty.
- **Input Controls:**
 - Action Dropdown: "Stay Out" or "Pit".
 - Target Lap Dropdown (if Pit is selected): Which exact lap in the current 5-lap block to pit (e.g., if in Round 2, options are Lap 6, 7, 8, 9, or 10).
 - New Compound Dropdown (if Pit is selected): Soft, Medium, or Hard.
- **Submit Button:** Once clicked, changes to a disabled "Choices Locked" state to prevent edits.

State B: Spectator Phase (Simulation Running)

- The input controls disappear or lock.
- **Status Banner:** "Simulation in Progress - Look at the Main Screen."
- **Live Telemetry Box:** Updates their total time and tire age lap-by-lap as the server processes the block.

2. ADMIN PAGE (The Director's Cut)

The Admin page is the master control room. It must have override capabilities in case a team's Wi-Fi drops or a bug occurs.

- **Session Controls:**
 - **Timer Button:** "Start 3-Minute Strategy Window." (Pushes the Submission Phase UI to all Team Pages).
 - **Execute Simulation Button:** "Run Next 5 Laps." (Locks all team inputs, triggers the math calculations on the backend, and pushes the animation to the Projector Page).
- **Submission Monitor:**
 - A live table showing all teams and a green checkmark when their submission is locked.
 - **Force Submit Button:** If the 3-minute timer expires and a team hasn't clicked submit, the admin clicks this to force them into a "Stay Out" default state.
- **God-Mode Overrides (Crucial for live events):**
 - Ability to manually edit a team's Total Time, Tire Age, or Compound (necessary if a team accidentally submits the wrong thing and you want to grant mercy, or if the system glitches).
- **Event Triggers:**
 - **Toggle Weather Alert:** Overrides standard formulas with Wet/Intermediate formulas for the next block.

3. MAIN RACE PAGE (The Big Screen / Projector View)

This is the only screen the audience watches during the 60-second simulation phase. It runs purely on read-only data sent from the backend.

- **The Leaderboard (Left Side - 40% of screen):**
 - Ranks P1 through Px based on Total Elapsed Time.
 - Must dynamically re-sort as lap times are calculated.
 - Shows intervals (e.g., P2 is + 3.4s behind P1).
 - Shows current tire icon and age next to each team name (e.g.,  S - 4).
- **The Track Visualizer (Right Side - 60% of screen):**
 - A looping 2D graphic or progress bar for each team showing relative positions.
 - **Visual Cues:** When a team takes a pit stop, their icon flashes in the "Pit Lane" zone, and their lap time visibly spikes by 20s, causing them to drop down the leaderboard instantly.
- **The Alert Banner (Top Center):**
 - Flashing text for dynamic events.
 - **Crucial Feature:** If the backend detects 4 or more teams pitting on the same lap, this banner must aggressively flash: "PITLANE CONGESTION: +6 SECOND PENALTY APPLIED TO TEAMS A, B, C, D".

4. BACKEND LOGIC & STATE MANAGEMENT (The Engine)

The technical team must build the backend to process the 5-lap block in a specific mathematical sequence before sending the results to the frontend.

The Processing Loop (When Admin clicks "Execute"):

1. **Read Intentions:** Pull every team's submission for the 5-lap block.
2. **Congestion Check (Game Theory Logic):**
 - The server groups all pit stops by the target lap number.
 - If `count(pit_lap_X) >= 4`, it updates the penalty variable for those specific teams for that specific lap from 20s to 26s.
3. **Calculate Lap by Lap:** For Lap 1 of the block, calculate the math for all teams based on their tire age. Update total time.
4. **Execute Pits:** If a team's target pit lap matches the current lap being calculated, add the penalty time and reset their tire age $x = 0$. Change their active compound string.
5. **Store & Broadcast:** Save the new data array into the database and broadcast the chronological updates via WebSockets to the Main Race Page so the visualizer can animate the changes lap-by-lap.

ADMIN PAGE (The Race Director Deck)

The Admin Page must now act as a live command center to map powers and change weather states *before* executing the next 5-lap block.

A. Global Track State Controls

- **Weather Dropdown:** Dry (Default) | Wet | Drying.
- *Action:* Changing this instantly unlocks Intermediate (Inter) and Full Wet compounds on the Team Pages and changes the backend calculation formulas.

B. Superpower Assignment Matrix

A table listing all teams with the following manual controls:

- **Power Dropdown:** Assigns an active power to a team for the upcoming block (None | Hammertime | Blitzkrieg | Rainmaster | Plan E | Minister of Defence).
- **Alonso Target Selector:** If Minister of Defence is selected, a secondary dropdown appears to select the *Target Team* to block.
- **Leclerc Override Box:** If Plan E is selected (after you roll the physical dice in the room), an input box appears to manually set that team's pit penalty to 5 or 30.
- **Apply Powers Button:** Locks in the modifiers to the backend state before the simulation runs.

2. BACKEND LOGIC (Python / FastAPI Updates)

The backend needs a robust modifier system. When the Admin maps a power, it temporarily alters the car's object properties for that specific 5-lap calculation.

A. New State Variables per Team:

```
JSON
{
  "team_id": "Team 1",
  "active_power": null,
  "is_pit_blocked": false,
  "base_pit_penalty": 20.0,
  "tire_age_frozen": false,
  "weather_immunity": false,
  "congestion_immunity": false
}
```

B. How the Powers Alter Backend Math:

When the Admin assigns a power, the backend maps it to these state changes for the current block:

- **Hammertime:** Sets `tire_age_frozen = true`. The calculation loop forces $x = 1$ for all 5 laps, ignoring actual tire age.
- **Blitzkrieg (Verstappen):** Sets `base_pit_penalty = 10.0` and `congestion_immunity = true`. They bypass the + 6s traffic check entirely.
- **Rainmaster (Schumacher):** Sets `weather_immunity = true`. If the track is `Wet` and they are on Slicks, they only add + 2.0s per lap instead of + 12.0s.
- **Plan E (Leclerc):** Replaces `base_pit_penalty` with the exact integer (5 or 30) typed in by the Admin.
- **Minister of Defence (Alonso):** Sets `is_pit_blocked = true` for the *Target Team*.

C. The Weather Formulas (Lap 26+):

The backend `calculate_lap_time()` method must check the Global Track State:

- **Wet Track Slicks:** $T(x) = DryFormula + 12.00$ (Unless `weather_immunity` is true).
- **Intermediate:** $T(x) = 84.00 + 0.80x$
- **Full Wet:** $T(x) = 87.00 + 0.30x$
- **Drying Track:** Multiply the degradation coefficient (x) of Inters and Wets by 3.0 (e.g., Inters become $0.80x \times 3 = 2.40x$).

3. TEAM PAGE (Student Dashboard Updates)

- **Weather Expansion:** When the Admin switches the track to `Wet`, the "New Compound" dropdown dynamically updates to include `Intermediate` and `Full Wet`.
- Team lap time must change when the power used on them

4. MAIN RACE PAGE (Projector Visualizer)

The frontend visualizer needs to highlight the chaos so the audience can see the powers in action.

- **Weather Overlays:** If Track State is **Wet**, render visual rain on the canvas/UI.
- **Power Indicators:** Next to a team's name on the live leaderboard, display a small icon if a power is active (e.g., a hammer for Hamilton, a shield for Alonso).
- **Sector Highlights:** If Hammertime is active, that team's lap times should flash purple (F1 color for fastest absolute sector) as they fly up the leaderboard.

RACE 1: THE SPRINT SHOOTOUT (15 Laps / 3 Blocks)

Track Environment: Permanent Dry

Special Rules: No Superpowers, No Weather Shifts, Pure Baseline Math.

1. Lap Time Formulas ($T(x)$ in seconds)

Where x is the current tire age in completed laps on that set ($x = 1, 2, 3, \dots$).

- **Soft Compound:**

- $$T_{soft}(x) = 78.00 + 0.15 \cdot x^2$$

- **Medium Compound:**

- $$T_{med}(x) = 80.00 + 1.80 \cdot x$$

- **Hard Compound:**

- $$T_{hard}(x) = 82.00 + 0.60 \cdot x$$

2. Pit Stop Penalties

- **Standard Pit Penalty (P):** + 20.00s added to the total time on the pit lap.
- **Tire Reset:** Pitting resets tire age $x = 0$ for subsequent calculations.
- **Pitlane Congestion:** If ≥ 4 teams pit on the exact same lap, add + 6.00s traffic penalty to each (Total pit cost = 26.00s).

RACE 2: THE MAIN GRAND PRIX (50 Laps / 10 Blocks)

Track Environment: Dynamic (Dry $\rightarrow$ Wet $\rightarrow$ Drying)

Special Rules: Weather Modifiers, Dynamic Tire Compounds, Pit Congestion, Admin Superpowers.

1. Global Track State Formulas

Phase A: Dry Track (Laps 1–25)

- **Soft:** $T(x) = 78.00 + 0.15 \cdot x^2$
- **Medium:** $T(x) = 80.00 + 1.80 \cdot x$
- **Hard:** $T(x) = 82.00 + 0.60 \cdot x$

Phase B: Wet Track (Laps 26–35)

When Global State is set to **Wet**:

- **Intermediate Tire (Green):**
 - $T_{inter}(x) = 84.00 + 0.80 \cdot x$
- **Full Wet Tire (Blue):**
 - $T_{wet}(x) = 87.00 + 0.30 \cdot x$
- **Dry Slicks on Wet Track (Soft / Medium / Hard):**
 $T_{\text{slick_in_rain}}(x) = T_{\text{dry_formula}}(x) + 12.00$
(Unless team has Schumacher Rainmaster power applied).

Phase C: Drying Track (Laps 36–50)

When Global State is set to **Drying**:

- **Dry Slicks (Soft / Medium / Hard):** Revert to standard dry formulas.
- **Intermediate Tire (Overheating in Dry):**
 $T_{\text{inter_drying}}(x) = 84.00 + 2.40 \cdot x$
(Degradation coefficient multiplied by 3.0 $\times$: $0.80 \times 3 = 2.40$)
- **Full Wet Tire (Overheating in Dry):**
 $T_{\text{wet_drying}}(x) = 87.00 + 0.90 \cdot x$
(Degradation coefficient multiplied by 3.0 $\times$: $0.30 \times 3 = 0.90$)

2. Pit Stop & Traffic Logic

Base Pit Penalty = 20.00s

```
IF count(teams_pitting_on_current_lap) >= 4:
    Pit Penalty = 26.00s (+6.00s Traffic Jam)
ELSE:
    Pit Penalty = 20.00s
```

3. Superpower Mathematical Overrides (Admin Mapped)

When the Admin assigns a superpower to a team for a 5-lap block, the backend applies these specific mathematical overrides:

Driver Power	Backend Parameter Override
Lewis Hamilton (<i>Hammertime</i>)	Force tire age $x = 1$ for all 5 laps in the calculation loop, regardless of actual tire age.
Max Verstappen (<i>Blitzkrieg</i>)	Set Base Pit Penalty = 10.00s and bypass the ≥ 4 pitlane congestion check (congestion_immunity = True).
Michael Schumacher (<i>Rainmaster</i>)	If on dry slicks during Wet track, change wet penalty from $\$+12.00\text{\textit{s}}$ to $\mathbf{\$+2.00\text{\textit{s}}}$.
Charles Leclerc (<i>Plan E</i>)	Replace Base Pit Penalty with the exact number manually typed by the Admin (5.00s on successful roll, or 30.00s on failed roll).
Fernando Alonso (<i>Minister of Defence</i>)	Set is_pit_blocked = True on the <i>Target Team</i> (disables pit execution for that block).

4. Backend Pseudo-Code for Lap Calculation

Python

```
def calculate_lap_time(car, track_state, is_pitting, lap_number, congestion_count):
    # 1. Determine Base Pace
     $x = 1$  if car.is_hammertime else car.tire_age
```

```

if track_state == "DRY":
    if car.compound == "SOFT": base_time = 78.00 + 0.15 * (x ** 2)
    elif car.compound == "MED": base_time = 80.00 + 1.80 * x
    elif car.compound == "HARD": base_time = 82.00 + 0.60 * x

elif track_state == "WET":
    if car.compound == "INTER": base_time = 84.00 + 0.80 * x
    elif car.compound == "WET": base_time = 87.00 + 0.30 * x
    else: # On dry slicks in rain
        slick_dry_time = calculate_dry_time(car.compound, x)
        wet_penalty = 2.00 if car.is_rainmaster else 12.00
        base_time = slick_dry_time + wet_penalty

elif track_state == "DRYING":
    if car.compound == "INTER": base_time = 84.00 + 2.40 * x
    elif car.compound == "WET": base_time = 87.00 + 0.90 * x
    else:
        base_time = calculate_dry_time(car.compound, x)

# 2. Add Pit Stop Penalty if Pitting on this Lap
pit_time = 0.0
if is_pitting:
    if car.is_blitzkrieg:
        pit_time = 10.00 # Verstappen flat 10s stop, immune to traffic
    elif car.is_plan_e:
        pit_time = car.plan_e_custom_penalty # 5s or 30s
    else:
        pit_time = 26.00 if congestion_count >= 4 else 20.00

# Reset car status for subsequent laps
car.tire_age = 0
car.compound = car.next_compound

car.tire_age += 1
return base_time + pit_time

```

NOTE: Admin can only toggle between track state(wet, dry)

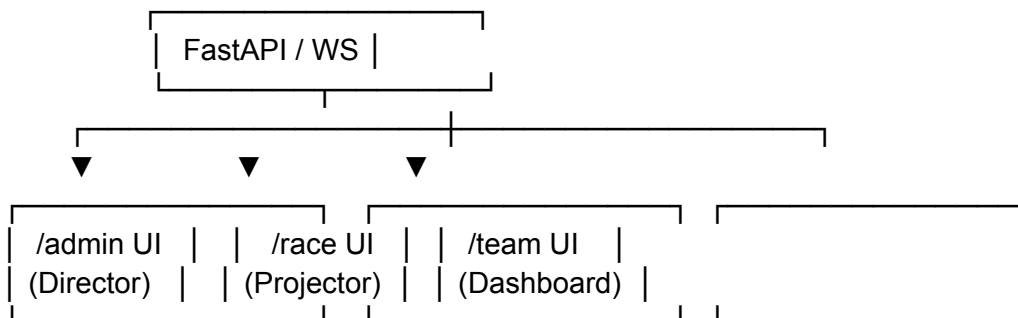
1. HARDWARE SETUP & DISPLAY ROUTING (Single Laptop Architecture)

You **do not need two separate computers**. A single laptop connected to the hall projector via HDMI is sufficient.

Recommended Display Modes

- Extended Desktop Mode (Best Operational Flow):**
 - Laptop Screen (Display 1 - Private):** Hosts the </admin> page. The host/admin can see pending submissions, active powers, override buttons, and live controls privately without the audience seeing behind-the-scenes adjustments.
 - Projector Screen (Display 2 - Public):** Displays </race> (The Big Screen Live Visualizer & Dynamic Leaderboard).
- Single-Screen Toggle Mode (Fallback):**
 - If running in duplicate/single-screen mode, the </admin> navbar must include a persistent toggle button: [[Switch to Projector View \(/race\)](/race)].
 - The </race> view must have a discrete return button: [[Return to Admin Panel \(/admin\)](/admin)] (password/PIN protected or hidden in the corner).

2. DETAILED PAGE SPECIFICATIONS



Page A: The Admin Control Deck (</admin>)

The Admin Page must provide complete transparency into team decisions and full manual override authority.

1. The Strategy Inspection Table (Submission Phase)

During the 3-minute strategy window, the Admin page displays a live grid of all connected teams:

Team Name	Current Status	Selected Action	Target Pit Lap	Target Compound	Declared Power	Override Action

Team 01	SUBMITTED (Green)	PIT	Lap 8	HARD	None	[Edit] [Block]
Team 02	PENDING (Yellow)	<i>Waiting...</i>	—	—	Hammertime	[Force Submit]
Team 03	SUBMITTED (Green)	STAY OUT	—	—	None	[Edit] [Block]

- **Real-Time Strategy Visibility:** The moment a team hits "Submit" on their dashboard, their exact choices populate on the Admin grid.
- **Espionage / Manual Edit Modal:** When a team uses a Radio Intercept token or an Admin needs to correct a bug, clicking [Edit] opens a modal to manually modify **Action**, **Pit Lap**, or **New Compound**.

2. Global State & Superpower Matrix

- **Track State Radio Buttons:** [(o) DRY] [() WET]
 - Toggling **WET** unlocks Intermediate/Wet options on Team dashboards and triggers wet calculation logic.
- **Superpower Mapper Table:**
 - **Team Selector:** Dropdown to assign powers (**Hammertime**, **Blitzkrieg**, **Rainmaster**, **Plan E**, **Minister of Defence**).
 - **Alonso Target Selector:** Dynamically appears if **Minister of Defence** is selected (choose target team to lock their pit button).
 - **Leclerc Die-Roll Input:** Dynamically appears if **Plan E** is selected (number input field to enter **5.0** or **30.0**).
- **Execution Trigger:**
 - [**RUN NEXT 5 LAPS**] Button: Locks submissions, runs the backend math loop, broadcasts state updates via WebSockets, and shifts **/race** into animation mode.

Page B: The Student Dashboard (/team)

1. Persistent Leaderboard & Interval Monitor

The Team page must include a collapsible or split-screen **Live Leaderboard** so teams can track competitors without relying exclusively on the main projector screen:

- **Position & Interval Delta:** Displays current rank (P1 to Px) and gap to the car ahead (e.g., **+1.850s**).

- **Tire Telemetry:** Shows each competitor's compound badge (● S, ● M, ● H, ● I, ● W) and current tire age (x).
- **Lap History Table:** Displays the team's exact lap-by-lap completion times for the current stint.

2. The 3-Minute Strategy Control Form

- **Tire Status Card:** Displays Current Compound, Current Tire Age (x), and degradation formula reference.
- **Action Selector:** Stay Out vs. Pit Stop.
- **Sub-Selections (Only active if Pit Stop is selected):**
 - **Pit Lap Dropdown:** Populated only with laps belonging to the current 5-lap block (e.g., in Block 2: Lap 6, 7, 8, 9, 10).
 - **New Compound Dropdown:** Soft, Medium, Hard (plus Intermediate and Full Wet if Admin has set track to WET).
- **Active Modifier Alerts:**
 - If targeted by Alonso: Banner displays ⚠ "MINISTER OF DEFENCE ACTIVE - PIT ENTRY CLOSED", and the Pit Stop option is disabled.
 - If Hammertime active: Banner displays 🛠 "HAMMERTIME ACTIVE - DEGRADATION COEFFICIENT FROZEN".

Page C: The Main Projector Screen (/race)

- **Left Panel (40% Width) - Dynamic Timing Tower:**
 - Chronologically re-sorts cars lap-by-lap as calculations stream in.
 - Flashes green sector icons for fastest laps and purple for active *Hammertime*.
- **Right Panel (60% Width) - 2D Track / Progress Visualizer:**
 - Renders car positions along the 5-lap loop.
 - **Pit Lane Zone:** When a car's lap time includes a pit stop penalty, the visualizer routes their marker through a dedicated pit lane graphic.
- **Global Alert Banner (Top Header):**
 - Displays weather alerts (TRACK WET - RAIN PROTOCOL ACTIVE).
 - Displays pitlane congestion warnings (⚠ CONGESTION: 4+ CARS PITTED ON LAP X -> +6.00s TRAFFIC PENALTY).

3. TECHNICAL EDGE-CASE SPECIFICATION & ENGINE LOGIC

Edge Case 1: The "Round 0" Initial Starting Grid State

- **Problem:** At lights out, the engine cannot compute Lap 1 without a starting compound assignment.
- **Engine Requirement:**

- Prior to Block 1, the Admin triggers a "**Grid Initialization Window**" (120 seconds).
- Teams must select their initial starting compound (**SOFT**, **MEDIUM**, or **HARD**).
- Backend initializes each car object at Lap 0 with **tire_age = 0** and **total_race_time = 0.000**.
- If a team fails to select, default fallback is automatically assigned as **MEDIUM**.

Edge Case 2: Exact In-Lap vs. Out-Lap Chronological Order

- **Problem:** Ambiguity around when tire wear applies versus when the new compound takes effect.
- **Mathematical Sequence (Example: Pitting on Lap 8 of Laps 6–10):**

Lap 6: $T_{old}(x=6)$

Lap 7: $T_{old}(x=7)$

Lap 8: [IN-LAP / PIT STOP]

Lap Time = $T_{old}(x=8) + \text{Pit_Penalty}$ (20.0s or 26.0s)

---> Tire reset executes at END of Lap 8: ($x := 0$, $\text{compound} := \text{new_compound}$)

Lap 9: [OUT-LAP]

Lap Time = $T_{new}(x=1)$

Lap 10: Lap Time = $T_{new}(x=2)$

Edge Case 3: Superpower Auto-Expiration & One-Time Lockout

- **Problem:** Modifiers remaining active across multiple blocks, causing state pollution.
- **Engine Requirement:**
 - **Block Reset:** At the completion of the 5-lap calculation block, all active boolean flags (**is_hammertime**, **is_blitzkrieg**, **is_rainmaster**, **is_plan_e**, **is_pit_blocked**) automatically reset to **False**.
 - **One-Time Lockout:** Every team object maintains a **has_used_power: bool** field. When any superpower is applied, set **has_used_power = True**. In subsequent rounds, that team's ID is disabled from the Admin Superpower assignment dropdown.

Edge Case 4: Track State Inversion (Transition from Wet to Dry)

- **Problem:** Teams remaining on wet tires when track conditions dry out.
- **Engine Requirement:**
 - The Admin sets the Global Track State to either **DRY** or **WET**.
 - When Global Track State is **DRY**:
 - If a car runs **Slicks**, compute standard dry formulas.
 - If a car remains on **Intermediate** tires: Apply overheating penalty formula:
$$T(x) = 84.00 + 2.40x$$
 - If a car remains on **Full Wet** tires: Apply overheating penalty formula:

$$\blacksquare T(x) = 87.00 + 0.90x$$

Edge Case 5: AFK / Network Disconnection Fallback

- **Problem:** Unsubmitted payloads blocking backend processing when timers expire.
- **Engine Requirement:**
 - When the 3-minute submission window timer hits 00:00, the server automatically marks all pending teams as SUBMITTED.
 - For any team without an explicit payload, inject the default fallback state:
 - JSON

```
{
  "action": "STAY_OUT",
  "pit_lap": null,
  "new_compound": null
}
```

-
-
- The calculation engine immediately executes without throwing null-pointer exceptions or halting the event.

4. DATA CONTRACT: WEBSOCKET / JSON SCHEMAS

A. Team Submission Payload (POST /api/strategy/submit)

JSON

```
{
  "team_id": "team_ferrari",
  "round_number": 2,
  "action": "PIT",
  "pit_lap": 8,
  "new_compound": "HARD"
}
```

B. Admin Block Execution Payload (POST /api/admin/execute-block)

JSON

```
{
  "block_number": 2,
  "start_lap": 6,
  "end_lap": 10,
  "track_state": "DRY",
  "active_modifiers": [
    {
```

```

    "team_id": "team_redbull",
    "power": "BLITZKRIEG"
  },
  {
    "team_id": "team_aston",
    "power": "MINISTER_OF_DEFENCE",
    "target_team_id": "team_ferrari"
  },
  {
    "team_id": "team_mercedes",
    "power": "HAMMERTIME"
  }
]
}

```

C. Live Telemetry Stream Object (Broadcasted to */race* and */team*)

JSON

```

{
  "lap_number": 8,
  "standings": [
    {
      "position": 1,
      "team_id": "team_redbull",
      "driver": "Max Verstappen",
      "compound": "HARD",
      "tire_age": 1,
      "last_lap_time": 92.000,
      "total_race_time": 684.500,
      "interval_to_leader": 0.000,
      "status": "TRACK",
      "active_power": "BLITZKRIEG"
    },
    {
      "position": 2,
      "team_id": "team_ferrari",
      "driver": "Charles Leclerc",
      "compound": "HARD",
      "tire_age": 0,
      "last_lap_time": 105.350,
      "total_race_time": 691.950,
      "interval_to_leader": 7.450,
      "status": "IN_PITLANE",
      "active_power": null
    }
  ]
}

```

```

],
"pitlane_congestion_alert": true,
"congested_teams": ["team_ferrari", "team_mclaren", "team_alpine", "team_williams"]
}

```

5. SERVER-SYNCHRONIZED TIMER & DECOUPLED EXECUTION FLOW

This requirement defines the exact state machine for the 3-minute strategy window. The timer must be dictated by the server (Server-Side Truth) to prevent local clock cheating, and the countdown must be decoupled from the actual simulation execution to give the Admin live-event flexibility.

A. The Timer State Machine

- **State 1: Standby (Post-Block):** After a 5-lap simulation finishes, all `/team` dashboards default to a read-only "Standby" state. Teams can view their telemetry, but all form inputs are disabled.
- **State 2: The Admin Trigger:** The countdown does **not** loop automatically. The Admin must manually click `[Start 3-Minute Strategy Window]` on the `/admin` page.
- **State 3: WebSocket Synchronization:** Clicking the start button sends a WebSocket broadcast to all connected clients.
 - A synced `03:00` countdown appears on the `/race` projector screen and all `/team` dashboards.
 - The form inputs (Action, Target Lap, Compound) instantly unlock for the teams.
- **State 4: The 00:00 Auto-Lock (Pencils Down):** When the server-side timer reaches `00:00`, it broadcasts a global `LOCK_INPUTS` event.
 - All `/team` UI dropdowns and buttons instantly disable.
 - Any team that did not successfully POST their payload is assigned the default fallback state (`STAY_OUT`).
- **State 5: The Decoupled Execution (The Intercept Window):** Reaching `00:00` **must not** automatically start the next 5-lap visualizer. The system holds all submissions in a pending state. This pause allows the Admin to process verbal "Radio Intercepts," manually edit team strategies, or map Superpowers. The simulation only begins when the Admin explicitly clicks `[Run Next 5 Laps]`.

B. Developer Implementation Note Do not use `setTimeout` or `setInterval` relying purely on the client's local browser clock for the lock-out mechanism, as teams could theoretically manipulate local time. The server must be the ultimate authority that rejects any POST request received after its internal 3-minute window closes.